Luphra MicroCover

Per-Action Insurance for Autonomous AI Agents

One-Line Pitch

Luphra lets AI agents buy insurance per action, priced in real time and paid through x402 on Algorand.

Problem

- AI agents are starting to spend money, call tools, send messages, book services, and make decisions for users and companies.
- Today, companies mostly choose between blocking risky actions or giving agents broad permissions.
- Static insurance premiums do not fit agent behavior because agent risk changes per action, context, budget, vendor, and token usage.
- There is no simple way to price, pay for, and audit coverage at the exact moment an autonomous action happens.

Insight

Agents should not only be allowed or denied. They should be insurable.

Solution

- Luphra intercepts a proposed agent action before execution.
- It scores the action based on context, amount, vendor, user policy, model confidence, and historical behavior.
- It generates a dynamic micro-premium for that specific action.
- The agent pays the premium via x402 on Algorand.
- Luphra returns a verifiable coverage receipt, then the agent proceeds.

Demo Scenario

1. A procurement agent wants to buy API credits from a new vendor.
2. Luphra receives the proposed action and scores the risk.
3. Luphra returns: risk score, premium, coverage limit, and reasons.
4. The agent pays the premium with x402 on Algorand TestNet.
5. Luphra issues a coverage receipt with an Algorand transaction ID.
6. The agent executes the purchase with coverage attached.

Why Algorand

- Per-action insurance needs tiny, high-frequency payments.
- Dynamic premiums only work if payment fees are low and predictable.
- Agent workflows need fast finality so actions do not wait for slow settlement.
- Algorand provides a clean audit trail for payment and coverage receipts.

Why x402

- x402 lets payment happen inside the HTTP request flow.
- The agent can request coverage, receive payment requirements, pay, and retry automatically.
- This removes subscriptions, manual billing, and separate checkout flows.

Product Surface

- **Risk API:** accepts proposed agent action and returns risk score.
- **Premium API:** returns dynamic premium and coverage terms.
- **Coverage API:** issues receipt after x402 payment.
- **Dashboard:** shows insured actions, premiums, receipts, and claims.

Hackathon Build

- Python FastAPI server for Luphra coverage API.
- Mock agent client that proposes risky actions.
- Dynamic premium calculator.
- x402 payment-gated coverage endpoint on Algorand TestNet.
- Simple receipt page or JSON response with coverage details and transaction proof.

Closing

As agents become economic actors, every autonomous transaction needs risk control. Luphra turns insurance from a static monthly product into a programmable, per-action primitive for the agent economy.